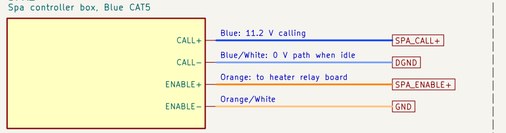
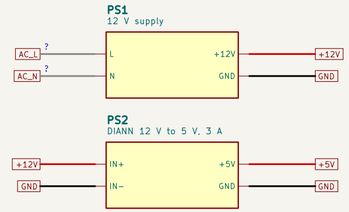
Solarthermal controller wiring (Waveshare ESP32-S3-POE-ETH-8DI-8RO)

Wire colour = conductor colour where documented. Pale tint = white-striped wire (Rob: in a pair, the wire with white is the ground). Grey = colour not documented, marked "?". Dashed = planned, not wired yet.

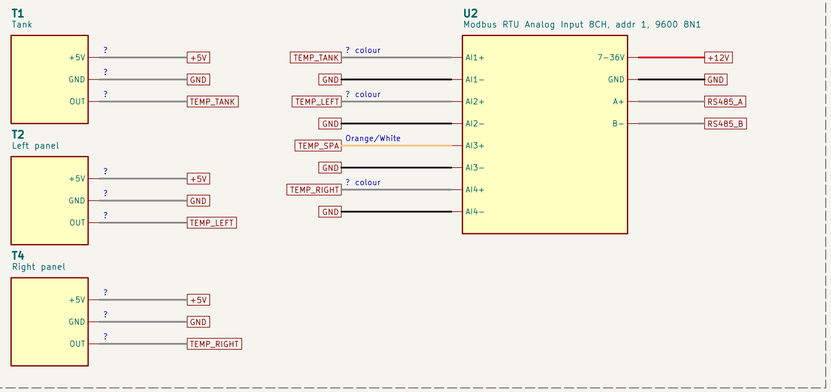
Spa (outdoors)



Box power

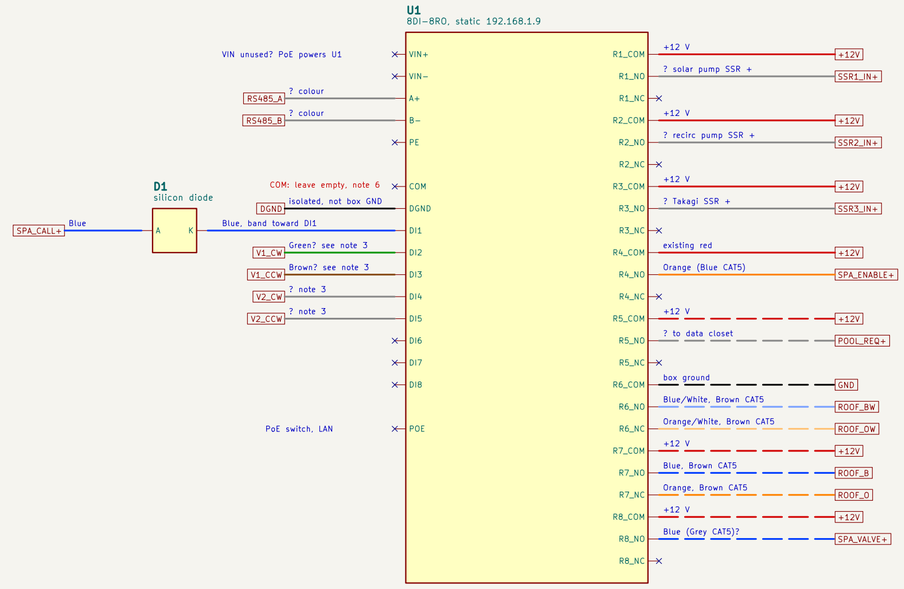


Temperature inputs (0–5 V, TMP36-type: (mV−500)/10 C)



U1

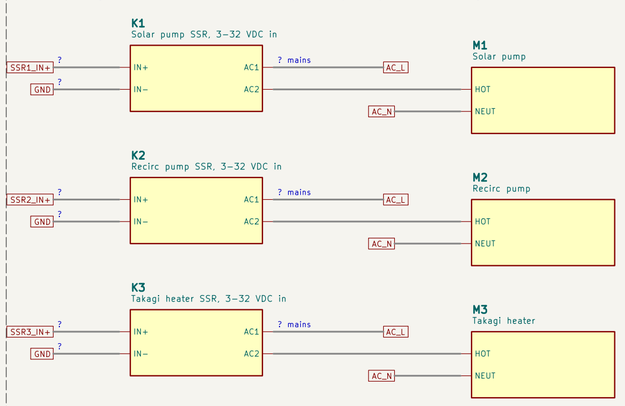
8DI-8RO, static 192.168.1.9



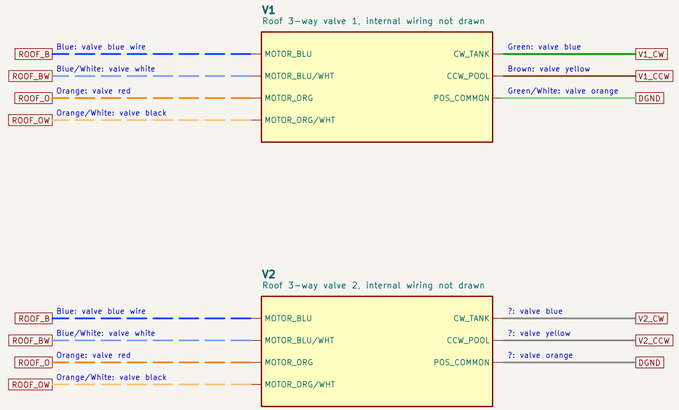
D1

silicon diode

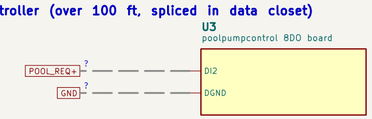
Mains side (SSRs switch the load; relays carry only SSR control)



Roof (both valves on one set of motor wires; each has separate motor-limit and position switches)



Pool controller (over 100 ft, spliced in data closet)



Notes

1. U1 relay wiring is from the proposed table and Rob's reports; "?" colours aren't documented.
2. R4: "existing red" and orange come from the proposed table; check on the wire.
3. DI2–DI5: Brown CAT5 labels give Green/Brown/Green-White for one valve. Valve 2's colours are unknown.
4. Valve internals and motor-limit wiring are deliberately not drawn.
5. Diode D1 stops idle current through the spa's low-resistance output.
6. Commons go on DGND. With them on COM, DI1–DI5 all read inactive.
7. R6 + R7 switch together: energized = pool, released = tank. Motor wires are Brown CAT5.

Tank = 1010 and pool = 0101 on DI2–DI5. Solar pump needs 1010; pool request needs 0101.
A commanded move not confirmed in 15 s sends an URGENT alert.

Drawn from the wiki, old sketch comments, cable labels and Rob's reports
Bedichek

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File: solarthermal.kicad_sch

Title: Solarthermal controller wiring

Size: User	Date: 2026-11-03	**Rev: 1**
KiCad E.D.A. 10.0.3		Id: 1/1